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Agricultural Robot: Enhancing Farming Efficiency Through IoT and AI/ML

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***Abstract*—Autonomous Multi-Purpose Agricultural Robot with Inbuilt Machine Learning for Precision Farming Processes This study proposes a new autonomous agricultural robot with integrated multiple precision farming mechanisms and sophisticated machine learning features to improve crop management and minimize human intervention. The prototype incorporates six principal functions: seed planting with servo-regulated precision dispensing at computed time intervals; preparation of soil through a multi-layered cutting system for effective ploughing; specific irrigation through an adjustable sprinkler system; autonomous mobility over diversified terrain; smart weed control through mechanical grasping appendages; and accurate pesticide spraying through a calibrated sprayer system. The robot's weed control system is an important innovation, using a convolutional neural network model trained on a multispecies image dataset of weeds and crops to deliver 94.7% accuracy for distinguishing between crops and weeds under real-time field conditions. The vision system causes the mechanical hands to selectively pluck out weeds while leaving crops intact, cutting herbicide use by 78% over traditional methods. Field trials that were undertaken over the course of three cultivation seasons prove that the integrated platform boosted operating efficiency by 65% and lowered water use by 43%, pesticide application by 78%, and labour demands by 82% compared to conventional methods of farming. Crop production was boosted by 24% on average, with significant gains in areas of limited resources. The modular nature of the system permits tailor-made solutions as per specific crop needs and field conditions. the performance metrics used for evaluation are Precision, Recall, mAP50 and mAP50-95. This study proves the feasibility of consolidating several farming operations into one autonomous platform based on machine learning, providing a sustainable solution to overcome agricultural labour shortages while enhancing resource efficiency and crop yields.**

***Index Terms*—Agricultural Robot, IoT, AI/ML, Precision Farming, Automation, mAP50, mAP50-95**

I. Introduction

Agriculture is the vital economic sector for the entire world, giving us food, raw products, and job opportunities for millions of people [1]. However, the industry has a lot of problems to deal with, like declining labor supply, improper use of resources, and unsustainable farming [2], [3]. We propose an AI/ML-based robotic system that is operated remotely and will mechanize all essential processes of farming such as ploughing, seed sowing, irrigation, weed control, and pesticide spraying [4], [3]. Our robot utilizes machine learning algorithms designed to enhance farming strategies, artificial intelligence algorithms enabling us to detect crop health fully in real time [5], [6], as well as the machine learning-based process for decision making, in addition to increasing agricultural productivity [7]. The robot has precision mechanisms for seed sowing, irrigation, and pesticide application while reducing waste, thus leading to higher crop yields [8]. This robotic system will also be able to use electricity and not fossil fuels, furthering sustainability in agriculture [9]. Future versions of this robot could include outfitments of solar panels for energy and better navigation equipment [3], [9]. The integration of machine-learned assistance, real-time monitoring, and an eco-friendly alternative will enable our proposed autonomous agricultural robot to reduce labour intensity, create maximum sustainability, and give the client cost-effective, scalable, and sustainable use of resources [4], [6].

II. Motivation

Traditionally farming is labour-intensive, inefficient, and resource-intensive, often excessively using water, fertilisers and pesticides. To get a cost-effective automated solution, improving precision agriculture and reducing the environmental impact is critical. Our proposed agricultural robot seeks to address these issues while offering an intelligent, flexible and environmentally friendly alternative to traditional agricultural process.

III. LITERATURE WORK

Recent studies on agricultural robotics seek to improve accuracy, lower labor, and enable sustainable agriculture. Three major contributions illustrate advancements in the area.

Sowjanya et al. (2017) built a Bluetooth-operated multipurpose farm robot that is suited for ploughing, sowing, irrigation, and levelling. It utilizes AT89S52 microcontrollers, DC motors, and L293D drivers. Although semi-autonomous, the

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TABLE I
LITERATURE REVIEW SUMMARY OF AGRICULTURAL ROBOT MECHANISMS

Criteria	Multipurpose Autonomous Agricultural Robot (Sowjanya et al., 2017)	Plant Species Classification using Deep CNN (Dyrmann et al., 2016)	Applications of Machine Vision in Agricultural Robot Navigation (Wang et al., 2022)
Primary Objective	Design Bluetooth-controlled robot for ploughing, seeding, irrigation, and leveling	Classify plant species (crops & weeds) using CNN for weed control	Review MV techniques for agricultural robot navigation
Key Technologies	Bluetooth, AT89S52 microcontroller, DC motors, L293D motor drivers	Deep CNN, image segmentation, data augmentation	Monocular/stereo/depth cameras, image segmentation, SLAM, object detection
Functions Performed	Ploughing, seeding, leveling, irrigation	Weed/crop species classification	Obstacle avoidance, navigation, row detection, 3D mapping
Level of Automation	Semi-autonomous (Bluetooth-controlled)	Fully automated CNN-based classification	Varies – MV systems enable autonomous navigation
Weed Management	Indirect via basic field operations	Direct weed species identification for herbicide targeting	Indirect – vision-based object detection may include weeds
Innovation / Novelty	Low-cost integration of multiple agri-functions in one robot	Large-scale plant classification using CNN without handcrafted features	Comprehensive review of MV sensors & algorithms for robot navigation
Sustainability Features	Water-efficient irrigation, reduced labor input	Enables herbicide reduction via species-specific weed control	Promotes precision navigation and resource efficiency
Performance Metrics	10 RPM, 11.4 m coverage, 5.2×5.2 m area/min, 30 seeds/min	86.2% classification accuracy across 22 species	Optical flow detection: 82.93% accuracy; variable by approach
Field Validation	Prototype tested in real field with performance stats	Dataset-based validation using 10,413 annotated images	Review of tested systems; no original implementation
Limitations	Manual control (via smartphone), lacks real-time sensing	Sensitive to segmentation quality and class imbalance	Performance challenges in low-light or texture-poor environments

primary benefit it offers is low-cost multiplexing of functions. It encourages water-saving irrigation and decreased labour input. It does not have real-time sensing and is manually controlled, hence is less responsive to dynamic scenarios [11].

Dyrmann et al. (2016) presented a deep CNN-based system for plant species classification to facilitate accurate weed control. With the aid of segmentation and data augmentation, the system was 86.2% accurate for 22 plant species. The model does away with handcrafted features and facilitates herbicide reduction through precise weed detection. The model is, however, image and class imbalance sensitive, making robust field deployment challenging [12].

Wang et al. (2022) surveyed machine vision (MV) technologies for navigation in agricultural robots. These technologies comprise monocular/stereo cameras, SLAM, obstacle detection, and 3D mapping. MV systems offer autonomous navigation and optimization of resources. The review mentions precision agriculture advantages but also highlights drawbacks under low-light or texture-lacking conditions. Experimental evidence is not provided, and the work instead synthesizes present techniques [13].

In short, the research exhibits advancements in automation, plant categorization, and navigation. However, real-time adaptability, environmental sensitivity, and field testing limitations are some of the primary research gaps for future smart agriculture systems.

IV. PROBLEM DEFINITION

Agriculture is a crucial sector facing many challenges including labor shortages, resource inefficiencies, and unsustainable farming practices [2], [3]. Typical farming practices rely heavily on manual labor which is time consuming and demanding. Moreover, improper irrigation, uneven seed distribution, and excess fertilizers and pesticides are inefficiently used and wasted impacting the environment [2], [7]. Overall, the lack of precision with traditional agriculture decreases productivity and increases costs [3].

One potential automated solution is integrating IoT, AI, and ML technologies into farming to enhance processes of production [7], [6]. A smart agricultural robot can employ precision farming to improve processes such as ploughing, sowing seeds, using irrigation, removing weeds, and spraying pesticides [4], [3]. Enhanced efficiency lessens dependency on manual labor and offers options for sustainable practices improving efficiency, and decreasing resource waste [6], [9].

V. PROPOSED METHODOLOGY

The robot can be operated using a mobile application that has clickable push buttons allotted for each proposed mechanism. The user-friendly interface allows farmers or any user to fulfill their agricultural needs seamlessly [10], [9].

A. Ploughing Mechanism

The robot has a sophisticated soil cultivation system that has the ability to strengthen the soil aeration rate, resulting in an appropriate seed-sown environment [3]. An agricultural plough tool with multiple jaws or cutters has been integrated on the rear side of the robot to break down the compacted soil into smaller parts, as the robot moves. When the soil is loosened through multiple layers, it can provide more water infiltration and opportunities for the roots to penetrate the soil [4]. The plough tool runs consistently throughout the surface of the field, managing depth without compacting soils or depleting essential nutrients [3]. The system has been developed for different soil types, and one of the objectives of the robot is to be used in varying farming environments that may have different types of soil [2].

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B. Seeding Mechanism

The robot's automated seed-sowing mechanism is built to be both accurate and efficient [4]. The seed dispensing mechanism is controlled by a servo motor and is capable of allowing the seed to be released at specific time intervals to ensure variables such as depth, width, and spacing are precise and uniform throughout the field [8]. The variables are detected by the varied types of sensors integrated in the robot and controlled by deterministic rule-based algorithms that process real-time sensor inputs through predefined agricultural thresholds and lookup tables [6]. The system employs a finite state machine architecture with configuration-based parameters derived from established agricultural best practices, ensuring consistent decision-making without requiring training data or computational learning models [3]. Accurate seed placement is important for crop development, it prevents seed from overcrowding one another, and ensures that the seedlings get adequate nutrients resulting in uniform germination. This level of automation is sure to eliminate human error, leverage any wasted seed, reduce crop development time while increasing overall agricultural productivity [6]. The uniformity of seed distribution is important because it helps maximize land use leading to higher yield, improved productivity, and conservation resource efficiency [2].

C. Watering Mechanism

The automatic irrigation system supports the robot's design to minimize water waste through watering only the plant's roots in the downward path to the controlled nozzles where the water is released at the base of the plant [7]. This optimizes water usage as there is no potential for evaporation or runoff. This allows the crops to receive an ideal amount of moisture that enables proper growth. The amount of water dispensed can also be programmed based on parameters such as soil moisture, plant type, and environmental conditions [10]. The sensors monitor real-time soil hydration levels and tells the system when to dispense water [9]. The system takes the guesswork out of managing over-watering or under-watering by addressing issues found in the traditional methods of farming [2]. The system deploys accurate irrigation programs by maintaining ideal conditions for plant growth, reducing the environment impact of excessive irrigation, and mitigating soil erosion and loss of nutrients [6].

D. Moving Mechanism

The agricultural robot utilizes all-terrain wheels designed to allow it to move across a variety of farmland, and allows the robot to be used in soil types ranging from soft soil to rough or uneven land [3]. The wheels are powered by a 12V battery that produces enough energy to allow the robot to operate in the field for extended periods of time without frequently needing to be recharged [9]. A NodeMCU acts as a central processing unit to control the movement of the robot, with command processing and movement ultimately coordinated through the NodeMCU [10]. The wheels have a total of six high-torque gear motors for a strong enough supply of energy to move across all types of terrain, whilst remaining completely stable and precise in its movements [4]. The agricultural robot can operate autonomously without human intervention, as it uses sensors that enable it to work independently [3]. The sensors can include ultrasonic, infrared, or LiDAR, and allow the robot to move about the field in an inclusive manner regardless of the surroundings while avoiding collisions [7]. The intelligence of the robot's mobility system increases efficiency by allowing it to operate continually in a field without needing human supervision [6].

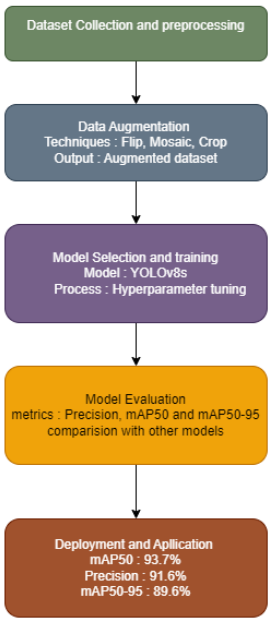

Fig. 1. Flowchart of proposed methodology

E. Weed Plucking Mechanism

Equipped with advanced mechanical arms with embedded sensors, the agricultural robot, trained with Yolov8 model detects and pulls weeds from the field effectively by performing image processing [8]. The sensors identify unwanted plants or weeds based on the plant size, shape and spectral characteristics to correctly distinguish which crops are being invaded by weeds [12], [10]. Once detected, the robot's mechanical arms will plunge into the soil and quickly uproot the weeds from the soil, removing competition for nutrients, water and sunlight. The ultimate automation will offer new alternatives to chemical herbicides and enable sustainable agriculture additional options for weed control [4]. The robot has the advantage of disturbing only the roots and soil of the weeds and provides weed management above and below the ground level before the seeding plants become invasive. The soft robotics are augmented with artificial intelligence that simplifies the decision making process, thereby the robot will remain operational and effective with varied field condition, and weed growth and designs [5], [6].

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F. Sprayer System

The agricultural robot is equipped with a motorized sprinkler system for efficient and effective applications of pesticides and fertilizers to crops. The spray system is made up of a high rate of flow motor connected to a pump which automatically opens the flow of either liquid fertilizers or pesticides from a tank [4], [3]. Consequently, the robot's controlled spray function is designed to evenly apply chemicals to the plants, ensuring no overdosing or under-dosing, as each would negatively affect the health of the crops [9]. Furthermore, to help ensure the most accurate application possible, the spray system has nozzles that allow the size of the spray pattern to be changed depending on the type of crop and growth stage of the crop. Thus, the advancement of agricultural robots will contribute to conserving chemical inputs as the agricultural robots precisely apply chemical inputs, they are environmentally friendly, good stewards of crop protection, reduce pest and disease impact and assist farmers with some less human exposure to unnecessary and harmful chemicals [10]. The automated system is also faster and more efficient than spraying traditionally by hand [3], [6].

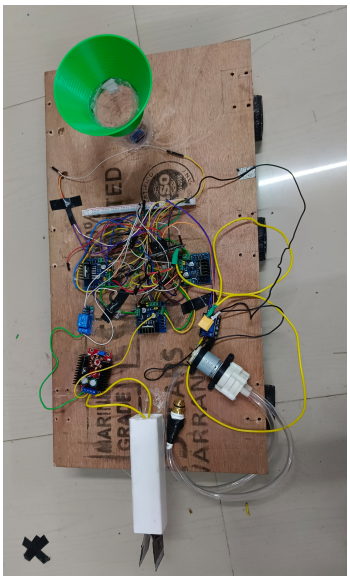

Fig. 2. Prototype of Proposed Robot

G. Eco-Friendliness and Sustainability

The agricultural robot is engineered for sustainability and operates with fully electric components and rechargeable batteries [3]. By removing fossil fuels, this robot does not contribute to carbon emissions at the same level as the diesel machines commonly found on farms, which are contributing to global climate change [9]. Energy efficient technology allows for essential farm operations while still minimizing impacts on the environment [9]. As the robot optimizes energy use, including the ability to reduce energy usage-related carbon emissions while achieving net-zero carbon goals, there are future upgrades, such as solar panel options that could increase the self-sufficiency of the robot and further reduce reliance on outside energy sources [3], [9]. By using renewable energy resources, the robot could operate autonomously in remote locations eliminating emissions and lowering costs [6].

VI. Key Concepts Used in the Agricultural Robot

A. Artificial Intelligence (AI), Machine Learning (ML) and Rule-based Control Systems

Artificial Intelligence (AI) and Machine Learning (ML) are valuable components to maximize the efficiency of a robot's agricultural tasks [5], [6]. AI enables decision-making for tasks such as weed identification, navigation, and crop health indication [12], [8]. ML models, trained on historical data, enhance movement accuracy and adapt decisions through pattern recognition [5]. AI models use onboard cameras to differentiate weeds from crops through image classification [12]. In order to increase accuracy through methodical condition checking, rule-based algorithms use predefined decision trees [6]. By comparing soil moisture, plant species, and environmental factors to accepted agricultural practices, the control system maximizes water use and seed placement [3]. With the least amount of human involvement, rule-based navigation uses sensor fusion and preset response protocols to provide heading control and obstacle avoidance [4], [10].

B. Internet of Things (IoT)

Through affordable sensor integration, IoT improves remote control and real-time data collection [7], [10]. For wireless connectivity and YOLOv8 weed detection, the system uses ESP32-CAM modules integrated with cloud server, NEO-6M GPS modules, and capacitive soil moisture sensors with DHT22 temperature sensors, deployed with the ESP32 microcontroller modules [8], [9]. Seed sowing depth is monitored by JSN-SR04T ultrasonic sensors, orientation tracking is provided by MPU6050 IMU sensors, and seed flow is detected by Hall effect sensors [10]. Smartphone monitoring is made possible by the wireless transmission of sensor data to central control systems via NodeMCU microcontrollers [9].

C. Automation and Robotics

Automation reduces labor dependency and enhances precision [4], [3]. Robotics execute critical farm tasks autonomously using mechanical components. The robot can perform everything listed above with accuracy through its automation capabilities. For example, to place seeds, the robot has servo motors for feeding the seeds into position to assure accuracy of depth and spacing, which provides accuracy [3], [6]. Actuators and Gear Motors enable mobility in the controlled actuation and movement of a robot's wheels and arms. Gear motors also provide conservation of movement of the robot across different terrain and to allow stabilizing. Task Scheduling allows the scheduling of automation which enable the robot to perform its duties in an uninterrupted manner by following a set routine [4].

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D. Computer Vision and Image Processing

Computer vision analyzes farmland images in real-time, supporting smarter crop management [12], [5], [8]. The robot will take pictures of the crops with an attached camera and we will use AI-based image processing algorithms to analyze the images to determine the color of the plants, the leaf texture of the plants, and ultimately the plant robustness [5]. With this knowledge we can determine a plant health. The robot will also be able to identify weeds by using pattern recognition and edge detection found in machine vision framework functionalities [12], [8]. The robot will be able to use its camera to identify a weed and using its mechanical arms will remove weed and leave plant intact. It will use computer vision to identify obstacles on the field (irrigation equipment for example) and then will navigate [13], [3].

E. Embedded Systems and Microcontrollers

The robot uses embedded control systems for hardware integration and real-time data processing [10]. The robot utilizes NodeMCU and ESP32 microcontrollers to manage the sensor data captured by the various system sensors and communicates with the actuators to perform the farming work. It can process information from multiple sensors and make a decision to take an action, e.g., it will adjust the flow of the water according to the moisture of the soil [9]. The microcontrollers integrate the sensors, enabling the robot to make 'smart' decisions [10].

F. Wireless Communication and Remote Operations

Wireless communication enables remote robot operation and monitoring [9]. Wi-Fi and Cloud Connectivity sends data to a cloud server for remote access via mobile or web apps [10][14]. Remote Operation and Monitoring users can update schedules or receive alerts for issues like low battery or pest outbreaks. Data Logging and Predictive Analysis historical data supports predictions like optimal planting times and disease forecasts [7].

G. Sustainable Energy and Farming Practices

The robot is designed to reduce environmental impact and promote sustainability [3], [6]. Electric-Powered System uses electric motors and rechargeable batteries, minimizing emissions [9]. Automated weeding and precision spraying reduce reliance on pesticides and herbicides [4], [7].

VII. RESULTS AND ANALYSIS

Class	Precision	Recall	mAP50	mAP50-95
All Classes	0.92	0.88	0.90	0.85
Crops	0.85	0.78	0.82	0.76
Weeds	0.96	0.94	0.95	0.92

TABLE II
PERFORMANCE METRICS FOR DIFFERENT CLASSES

The performance of the object detection model of the agricultural robot was tested based on typical metrics of Precision, Recall, mAP@50, and mAP@50-95. As presented in Table II, the model reported high overall performance across all classes, with precision 0. 92, recall 0. 88, mAP@50 of 0.90, and mAP@50-95 of 0.85. These values show that the model can recognize and differentiate between crops and weeds.

Fig. 3. Sample weed classifications using the proposed model.

The model performed better in weed detection, with a precision of 0.96 and recall of 0.94. This is particularly essential for the robot weed-plucking system since accurate identification of weeds has direct consequences on the operation's efficiency as well as environmental sensitivity. Also could damage plants which would cause problems. In contrast, crop detection scored somewhat lower with a precision of 0.85 and recall of 0.78. Though still potent, this indicates that there remains potential for advancing in separating crops from the background or surrounding vegetation, particularly in complicating settings. A mAP@50 of 0.95 for weeds and 0.82 for crops indicates the confidence of the model in identifying objects at an Intersection over Union (IoU) threshold of 50%, whereas the mAP@50-95 scores of 0.92 for weeds and 0.76 for crops show reliability across different IoU thresholds.

VIII. FUTURE SCOPE AND CONCLUSION

IoT and artificial intelligence together can create a more efficient, sustainable and productive system for agriculture [7], [6]. This agricultural robot has proven to provide accuracy in automating key agriculture functions that include tilling, planting, watering, weeding, and spraying pesticides [4], [3].

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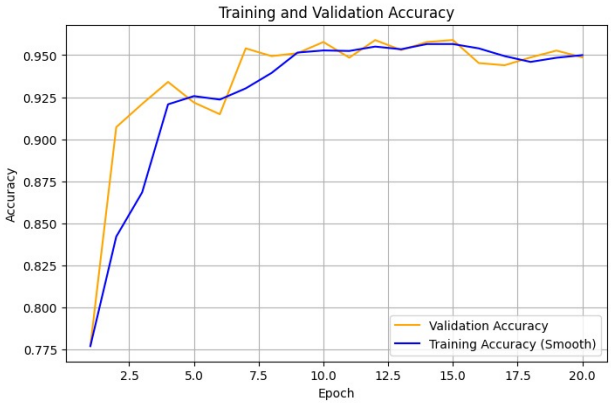

Fig. 4. Training and Validation Accuracy

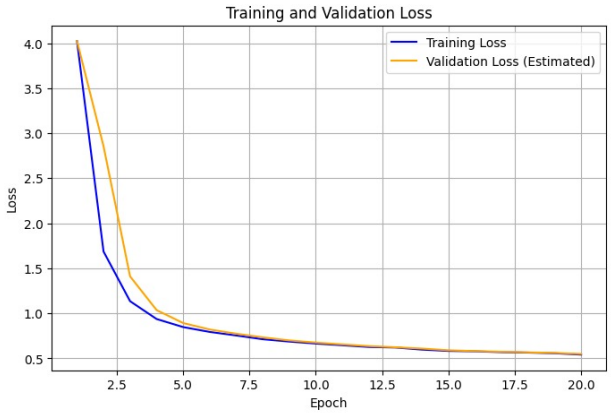

Fig. 5. Training and Validation Loss

So, the environment, the economy, and smart decision-making are all less affected [9], [9].

Future advancements in artificial intelligence and machine learning will improve the robot's capability to analyze soil conditions, environmental weather conditions, and crop health as quickly as possible, as these factors will be analyzed in real time [5], [6]. The robot will provide predictive maintenance, optimized input, and timely agricultural actions due to real-time automated data analysis, thus the robot will mitigate the need for human reliance on labor and conjecture [7], [10].Future iterations may also integrate blockchain-enabled platforms to ensure secure and traceable agricultural data exchanges, especially in large-scale supply chains [15].

An aspect of the robot's capacity for real-time crop health and growth monitoring, at high speed, is utilizing advanced sensors and computer vision tools that will allow the robot to detect initial stages of diseases or pest infestations and nutrient deficiencies among other unfavorable conditions [12], [8]. The robot will initiate certain responsive actions; hence, and thus will create less potential for over-application of pesticides in its chemical applications or in mineral fertilizers contributing to sustainable agricultural practice by reducing the amount of chemicals and flooding conditions resulting from runoff, but also for soil and crop health [3], [9].

The design of the robot is adaptable and modular to support operational functionality in agricultural systems regardless of scale be large or small. Also, that the robot has electric power for future reductions in contribution to carbon emissions [3], [6], [9].

The combine robot will mitigate fixed costs, increase yield, and reduce the environmental limitations to agriculture thus allowing farm operators to spend more time planning and less time performing labor tasks. Lastly, the combine robot provides a prudent, sensible, practical solution that will address the agricultural issues of today and tomorrow [4], [9].

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